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>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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TAM / SAM / SOM ANALYSIS

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NARRATIVE SUMMARY

Total Addressable Market (TAM)

The total potential global market for green hydrogen is poised to grow significantly, reaching 20-30 million tons annually by 2035 (Year 10 in our projection model). As cost parity with fossil-based hydrogen becomes a reality — even at electricity rates below $0.138/kWh — the total addressable market expands rapidly across multiple sectors and regions.

Serviceable Available Market (SAM)

Given Tobe Energy's ability to produce hydrogen at or below SMR-based costs (**$1.81/kg LCOH** at $0.10/kWh), an increasingly substantial share of the global market becomes accessible sooner than anticipated. As industrial consumers, such as ammonia plants, refineries, and steelmakers, are incentivized to switch to green hydrogen for both economic and environmental reasons, Tobe's regionally accessible slice of the market (on the order of **500,000–1,000,000 tons annually** in initial service regions) opens up, defining the serviceable available market.

Serviceable Obtainable Market (SOM)

By starting small — initially producing thousands of tons per year in Tulsa, OK — and consistently adding capacity every three months, Tobe Energy leverages its cost advantage, high efficiency (**>92% system efficiency** (HHV)), and stack durability to secure offtake agreements and steadily scale up production. While capturing even a few percentage points of its SAM (tens of thousands of tons relative to a half-million-ton regional market) may seem modest, this still translates into meaningful revenue and a strong competitive

foothold. As additional facilities mirror the Tulsa blueprint across the country, Tobe can incrementally grow its share, positioning itself as a significant player in the evolving green hydrogen economy.

ASSUMPTIONS AND CONTEXT

Market Timing

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- "Year 0" = Company's commercial start date (Q4 2026 per company context)
- 2-year ramp-up before initial production in Tulsa, OK
- By "Year 2," Tobe Energy reaches its initial **2,500 kg/day** capacity
- Adds 2,500 kg/day every 3 months (totaling an additional 10,000 kg/day per year)
- Over 8 more years (Years 3 to 10), capacity scales linearly

Electricity Costs and Production Costs

- Tobe Energy's cost structure allows hydrogen production near or below current SMR parity if electricity is cheaper than ~$0.138/kWh
- In Oklahoma, with favorable rates and partial solar integration, achieving ~$0.10/kWh is plausible, pushing production costs toward ~**$1.81/kg LCOH** (matching company context)
- This cost-competitiveness strongly positions Tobe Energy in early-adopter markets, particularly industries and applications seeking to decarbonize cost-effectively

Geographic Focus

- Initial production in Tulsa, OK: central location with access to industrial clusters (fertilizer plants, refineries, chemical producers) and potential hydrogen blending markets
- Future expansions "across the country" imply replication of the Tulsa model in multiple regions as adoption grows

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PRODUCTION CAPACITY GROWTH

YEAR	DAILY CAPACITY (KG/DAY)	ANNUAL PRODUCTION (TONS/YR)	NOTES
2	2,500	~912.5	Initial commercial operation
3	12,500	~4,562.5	+10,000 kg/day added
4	22,500	~8,212.5	+10,000 kg/day added
5	32,500	~11,862.5	+10,000 kg/day added
6	42,500	~15,512.5	+10,000 kg/day added
7	52,500	~19,162.5	+10,000 kg/day added
8	62,500	~22,812.5	+10,000 kg/day added
9	72,500	~26,462.5	+10,000 kg/day added
10	82,500	~30,112.5	Full-scale operation

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By Year 10, one steadily growing Tobe Energy facility produces ~30,000 metric tons/year

MARKET ANALYSIS

Total Addressable Market (TAM)

Global Context by Year 10:

- By the mid-2030s, global hydrogen demand could approach ~100-120 million tons/year, with green hydrogen capturing an accelerating share due to decarbonization efforts
- Assume ~20-25% of this is green hydrogen by Year 10: **~20-30 million tons/year globally** of green hydrogen (TAM)

U.S. Share of TAM:

- The U.S. might represent ~15% of global hydrogen demand
- If 20-25% of U.S. hydrogen is green by Year 10, and total U.S. hydrogen demand is ~10-15 million tons, green hydrogen could be ~2-4 million tons/year
- A plausible U.S. TAM slice by Year 10: **~3 million tons/year** of green hydrogen

TAM Summary:

- TAM (Global): **20-30 million tons/yr**
- TAM (U.S.): **~3 million tons/yr**

Serviceable Available Market (SAM)

Defining SAM: The SAM is the portion of the TAM that Tobe Energy can realistically target given market access, policy incentives, infrastructure, and customer readiness. Factors include:

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- Access to industrial clusters ready to switch to green hydrogen at ~$1–$1.50/kg
- Growing state-level incentives (e.g., hydrogen hubs, IRA tax credits)
- Regional pipelines and trucking logistics to move hydrogen cost-effectively

Regional & Sector Focus by Year 10:

- Fertilizer (ammonia) and refinery sectors in the Midwest/Plains region: prime candidates as they already consume large volumes of H_2 and face increasing decarbonization pressure
- Steelmaking and chemical feedstocks: as DRI (Direct Reduced Iron) plants and petrochemical facilities adopt green H_2, demand grows regionally

If we assume that by Year 10, about **500,000–1,000,000 tons/year** of green hydrogen demand emerges in the central U.S. region (including Oklahoma, surrounding states, and connected industrial corridors) that Tobe can physically serve, that sets our SAM at ~0.5–1.0 million tons/year regionally. As Tobe expands across the country, similar regional pockets appear. With multiple installations following the Tulsa blueprint, Tobe's SAM could approach the full U.S. green H_2 market (~3 million tons) if it achieves national presence.

SAM Summary:

- Conservative initial estimate: **~500,000–1,000,000 tons/year** in Tobe's initial service regions by Year 10
- Potential national SAM: **~3 million tons/year** if replication is successful

Serviceable Obtainable Market (SOM)

Tobe's Share of the SAM:

- By Year 10, one Tobe installation produces ~30,000 tons/year
- Relative to a 500,000–1,000,000 ton SAM, that's **3–6% market share** from one facility
- This is quite significant for a single plant, especially given that major industrial players will source from multiple suppliers
- If Tobe replicates this expansion model across 3–4 geographically strategic locations by Year 10 (each following a similar scaling pattern), total production could reach ~90,000–120,000 tons/year

- This would capture **9–24%** of a 500,000-ton SAM or **~3–12%** of a 1,000,000-ton SAM

Actual achievable share depends on investment, partnerships, and offtake agreements. Realistic SOM by Year 10 with one major production hub: ~3–6% of a 500,000–1,000,000 ton SAM. With multiple hubs: 10%+ of that SAM is feasible.

CONNECTING COSTS AND EFFICIENCY TO MARKET SHARE

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Cost Competitiveness

- At ~**$1.81/kg LCOH** (achieved with $0.10/kWh electricity), Tobe undercuts many existing green H_2 producers and competes with SMR hydrogen (after factoring in carbon costs or clean hydrogen incentives)
- High efficiency (**>92% system efficiency** (HHV), **94.7% stack efficiency** (LHV)) and durability reduces OPEX and downtime, improving profitability and enabling quicker capacity expansions and market penetration

Market Leverage

- Early-mover advantage in Oklahoma and nearby states
- As policies push green hydrogen adoption, Tobe can sign long-term offtake agreements with ammonia producers, refineries, and large industrial consumers who want secure, low-cost, zero-carbon hydrogen
- Incremental expansions every 3 months allow Tobe to respond dynamically to market demand, maintaining a strong position and potentially outgrowing initial SOM targets

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SUMMARY TABLE

METRIC	VALUE BY YEAR 10	NOTES
Global TAM (Green H_2)	~20-30 million tons/yr	Broad global decarbonization scenario
U.S. TAM (Green H_2)	~3 million tons/yr	Approx. 20-25% U.S. H_2 as green
SAM (Regional/Accessible)	500,000-1,000,000 tons/yr	Industrial corridors near Tobe's operations and potential expansions
Tobe's Production (Single Hub)	~30,000 tons/yr	Incremental expansions total ~82,500 kg/day by Year 10
SOM (Single Hub)	~3-6% of SAM	Depending on SAM size and market penetration
SOM (Multiple Hubs)	Potentially ~10%+ of SAM	If Tobe replicates the Tulsa model across several regions

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CONCLUSION

By Year 10, Tobe Energy can grow from 2,500 kg/day to over **82,500 kg/day** of production, translating to roughly **30,000 tons/year**. In a scenario where the U.S. green hydrogen market is maturing and the regional SAM is around 500,000–1,000,000 tons/year, this single Tobe facility can secure a meaningful market share (3–6%), indicating strong positioning.

Expanding this model to multiple locations across the country would allow Tobe Energy to scale its SOM further, potentially capturing a double-digit percentage of the regional SAM and establishing itself as a competitive leader in the burgeoning green hydrogen economy. With our **$1.81/kg LCOH** target and **>92% system efficiency** (HHV), Tobe Energy is uniquely positioned to capture this opportunity while raising the **$10M seed round** needed for initial deployment.